

Average Value, Accumulated Change, and Motion

Answer Key

AP Calculus AB/BC Applications of Integration

Quick Answers

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|-------|-----------------------|--------------------|
| 1. 3 | 5. 21.82 | 9. 42, 62 |
| 2. 65 | 6. 2.4, 11.6 | 10. 3, 5, 7.667, 1 |
| 3. -9 | 7. 1, 3, 2.667, 0.889 | |
| 4. 23 | 8. 4 | |
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Curriculum

Level: AP Calculus AB/BC Applications of Integration

FUN-6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 18 tagged checks.

1. Average value of $f(x) = x^2$ on $[0, 3]$.

Step 1. The question asks for an average value, so the setup is $\frac{1}{b-a} \int_a^b f$, with $a = 0$ and $b = 3$. Dividing by the interval length is what turns an accumulated area into a height.

Step 2. $\frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_0^3 = \frac{1}{3} \cdot 9 = 3.$

Step 3. Sanity check: f runs from 0 up to 9 on this interval, so an average of 3 is plausible — and it is below the midpoint value because x^2 spends most of the interval small. 3

2. Water entering a tank at $r(t) = 4t + 3$ litres per minute.

Step 1. The given function is a *rate*, and the question asks how much water arrives — that is accumulated change, so integrate the rate over $[0, 5]$.

Step 2. $\int_0^5 (4t + 3) dt = [2t^2 + 3t]_0^5 = 50 + 15 = 65.$

Step 3. The units check out: litres per minute times minutes gives litres. 65 L

3. Net displacement for $v(t) = 3t^2 - 12$ on $[0, 3]$.

Step 1. “Net displacement” means the signed change in position, so integrate the velocity itself — no absolute value.

Step 2. $\int_0^3 (3t^2 - 12) dt = [t^3 - 12t]_0^3 = 27 - 36 = -9.$

Step 3. The answer is negative because the particle spends the first two seconds moving backwards and does not fully recover. It ends up 9 metres behind where it started. -9 m

4. Total distance for $v(t) = 3t^2 - 12$ on $[0, 3]$.

Step 1. Total distance is the integral of *speed*, so first find where the velocity changes sign: $3t^2 - 12 = 0$ gives $t = 2$ in $[0, 3]$.

Step 2. On $[0, 2]$ the velocity is negative and on $[2, 3]$ it is positive, so split there: $\int_0^2 (3t^2 - 12) dt = 8 - 24 = -16$ and $\int_2^3 (3t^2 - 12) dt = (27 - 36) - (8 - 24) = 7$.

Step 3. Add the absolute values: $|-16| + |7| = 23$. Compare with problem 3: the same velocity function, the same interval, and a completely different number, because displacement lets the backward motion cancel and distance does not. 23 m

5. Average greenhouse temperature over $[0, 12]$.

Step 1. Average value again, this time of a sinusoidal model: $\frac{1}{12} \int_0^{12} T(t) dt$.

Step 2. $\int_0^{12} \left(18 + 6 \sin \frac{\pi t}{12} \right) dt = \left[18t - \frac{72}{\pi} \cos \frac{\pi t}{12} \right]_0^{12} = 216 + \frac{144}{\pi}$.

Step 3. Divide by the interval length 12: the average is $18 + \frac{12}{\pi} \approx 21.8197$, which rounds to 21.82. Note that it is above the baseline 18, as it must be: the sine term is non-negative across this whole half period. 21.82 degrees

6. Fuel remaining after four hours.

Step 1 (part a). The consumption rate is $c(t) = 0.4 + 0.1t$, so the fuel consumed is the accumulated change $\int_0^4 (0.4 + 0.1t) dt = [0.4t + 0.05t^2]_0^4 = 1.6 + 0.8 = 2.4$ litres. 2.4

Step 2 (part b). An accumulated change is only half the story — the amount at the end is the starting amount plus (here minus) that change: $14 - 2.4 = 11.6$.

Step 3. Sanity check: the tank started nearly full and the rate never exceeds 0.8 litres per hour, so losing under three litres in four hours is reasonable. 11.6 L

7. $v(t) = t^2 - 4t + 3$ on $[0, 3]$: rest times, distance, average speed.

Step 1 (part a). At rest means $v = 0$: factor $t^2 - 4t + 3 = (t - 1)(t - 3)$, so the particle is at rest at $t = 1$ and $t = 3$ seconds. $t = 1$ and $t = 3$

Step 2 (part b). Only $t = 1$ is interior, so split there. The antiderivative is $\frac{t^3}{3} - 2t^2 + 3t$. On $[0, 1]$ the integral is $\frac{4}{3}$; on $[1, 3]$ it is $0 - \frac{4}{3} = -\frac{4}{3}$. Total distance $= \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$ metres. $\frac{8}{3} \approx 2.667$

Step 3 (part c). Average speed is total distance over elapsed time — an average value applied to speed, not to velocity: $\frac{8/3}{3} = \frac{8}{9} \approx 0.889$ metres per second. (The average *velocity* would be 0, since displacement over $[0, 3]$ is zero.) 0.889

8. Average velocity of a cyclist with $v(t) = 3\sqrt{t}$ on $[0, 4]$.

Step 1. Average value of the velocity function: $\frac{1}{4} \int_0^4 3\sqrt{t} dt$. Rewrite the root as a power first.

Step 2. $\int_0^4 3t^{1/2} dt = [2t^{3/2}]_0^4 = 2 \cdot 8 = 16$, so the average is $\frac{16}{4} = 4$.

Step 3. Here the velocity never changes sign, so the displacement 16 metres equals the total distance and the average velocity equals the average speed. That agreement is what fails in problem 7. 4 m/s

9. Tank with water entering at $10 + 2t$ and draining at $3t$ litres per minute.

Step 1 (part a). Two rates act at once, so the change in volume is the integral of the *net* rate, in minus out: $(10 + 2t) - 3t = 10 - t$ litres per minute.

Step 2. $\int_0^6 (10 + 2t - 3t) dt = \left[10t - \frac{t^2}{2} \right]_0^6 = 60 - 18 = 42$ litres gained. 42

Step 3 (part b). Amount at the end = starting amount + accumulated net change = $20 + 42 = 62$. The net rate $10 - t$ stays positive on the whole interval, so the volume rises throughout — consistent with a gain. 62 L

10. Synthesis: $v(t) = 4 - t^2$ on $[0, 3]$ with $s(0) = 2$.

Step 1 (part a). Net displacement is the signed integral of velocity: $\int_0^3 (4 - t^2) dt =$

$\left[4t - \frac{t^3}{3} \right]_0^3 = 12 - 9 = 3$ metres. 3

Step 2 (part b). Position is the initial position plus the displacement, which is the Fundamental Theorem written for motion: $s(3) = 2 + 3 = 5$ metres. 5

Step 3 (part c). The velocity is zero at $t = 2$ and changes sign there, so split. On $[0, 2]$ the integral is $8 - \frac{8}{3} = \frac{16}{3}$; on $[2, 3]$ it is $3 - \frac{16}{3} = -\frac{7}{3}$. Total distance = $\frac{16}{3} + \frac{7}{3} = \frac{23}{3}$ metres.

$\frac{23}{3} \approx 7.667$

Step 4 (part d). Average velocity is the average value of v : $\frac{1}{3} \int_0^3 (4 - t^2) dt = \frac{3}{3} = 1$ metre per second. 1

Manual review (part e): the average speed is $\frac{23/3}{3} = \frac{23}{9} \approx 2.556$ metres per second, larger than the average velocity of 1. Accept any paragraph making the point that average velocity divides the *signed* displacement by the elapsed time, so the backward motion after $t = 2$ cancels part of the forward motion, while average speed divides total distance, where nothing cancels. A student who notes that the two agree exactly when the velocity never changes sign has the idea.